PUPPETMASTER- CONTROLLED DEEPFAKE GENERATION
ATTRIBUTE-AWARE AND IDENTITY-STABLE FACE SYNTHESIS

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ABSTRACT

We present **PuppetMaster**, a memory-efficient face attribute transfer framework that integrates Basel Face Model (BFM), a 3D Morphable Model (3DMM) conditioning into a Stable Diffusion-based latent diffusion pipeline, drawing inspiration from the DiffSwap paradigm. Unlike prior lightweight approaches that rely on heuristic landmark geometry or scalar approximations for 3D face attributes, our method extracts geometrically grounded BFM coefficients directly from 2D facial landmarks via a scipy-based closed-form fitting procedure that requires no pre-trained BFM regression network or external model weights. These coefficients are injected into a Stage-2 `PoseExpressionAdapter` MLP alongside a 512-dimensional ArcFace identity embedding and CLIP-derived disentangled features, yielding a 56-dimensional conditioning signal that modulates the UNet cross-attention layers through Adaptive Instance Normalization (AdaIN). At inference, source shape coefficients are composited with target pose and expression to achieve identity-preserving, pose- and expression-accurate face transfer, with a raised injection weight of 0.02 reflecting the improved geometric fidelity of BFM-grounded conditioning. Quantitative evaluation on the Labeled Faces in the Wild (LFW) dataset using ArcFace identity similarity and SSIM metrics, together with ablation studies over per-attribute conditioning weights (identity λ_{id}, pose λ_{pose}, expression λ_{expr}, shape λ_{shape}), demonstrates that BFM conditioning yields improved disentanglement over landmark-only baselines.

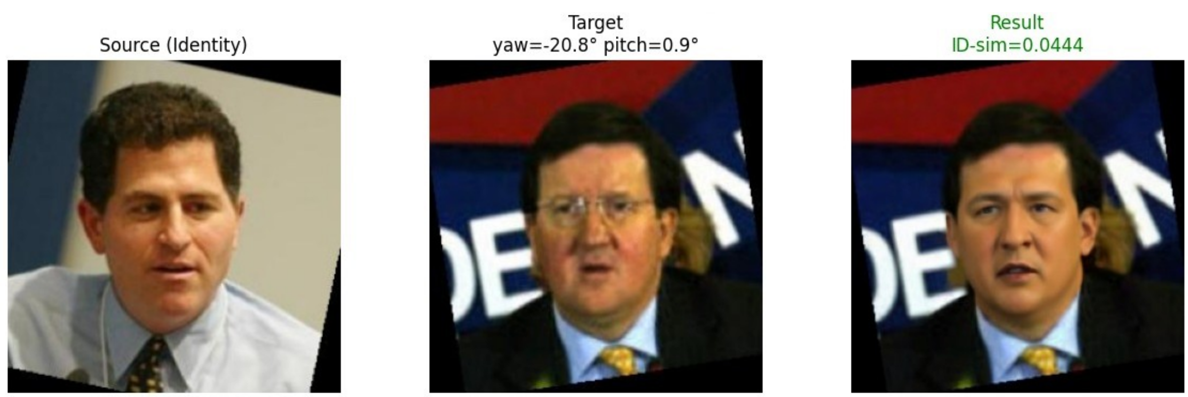

Figure 1: Sample Output of PuppetMaster

1 Introduction

Face attribute transfer—the task of synthesising an image that preserves the identity of a source subject while adopting the pose, expression, or illumination of a target—is a long-standing problem in computer vision with applications spanning entertainment, virtual reality, security, and human-computer interaction. Early approaches relied on 2D warping and blending techniques, but these methods are notoriously brittle under large pose variations and struggle to preserve fine-grained identity details. The emergence of Generative Adversarial Networks (GANs) significantly advanced the state of the art; however, GAN-based pipelines are sensitive to training instability and frequently suffer from mode collapse and identity leakage—a phenomenon in which target-face identity cues inadvertently overwrite those of the source.

More recently, *Latent Diffusion Models* (LDMs) have demonstrated remarkable image synthesis quality by operating in the compressed latent space of a pre-trained Variational Autoencoder (VAE), effectively decoupling perceptual compression from the generative denoising process. Their iterative, stochastic refinement procedure yields outputs of exceptional realism and diversity, and their conditioning mechanisms—most notably cross-attention injection and Adaptive Instance Normalisation (AdaIN)—provide natural hooks for fine-grained attribute control. Despite these advantages, naively adapting a pre-trained LDM for face attribute transfer remains non-trivial: the model must simultaneously respect a high-dimensional identity constraint (typically a 512-dimensional ArcFace embedding) and a structured 3D geometric constraint governing head pose and facial expression, all while operating within the tight memory budget imposed by consumer-grade hardware.

A key insight from *DiffSwap* is that disentangling identity from dynamics requires a geometrically grounded representation: if shape, pose, and expression are encoded as meaningful 3D coefficients rather than flat 2D heuristics, the conditioning signal can be cleanly decomposed into *what a face looks like* (identity shape) and *how it is animated* (target pose and expression). The Basel Face Model (BFM), a principal component analysis-based 3D Morphable Model (3DMM) fitted to a large corpus of 3D face scans, provides precisely such a representation: a low-dimensional linear subspace whose axes correspond to interpretable axes of facial variation, separable into shape (identity geometry) and expression (dynamic deformation) components.

The core inference strategy follows the DiffSwap paradigm: at inference time, source BFM shape coefficients are composited with target pose and expression coefficients before conditioning, so that the generated face inherits the three-dimensional bone structure of the source subject while adopting the head orientation and facial animation of the target. Face segmentation is provided by either a pre-trained BiSeNet (when available) or a fast landmark-derived ellipse mask fallback, and identity similarity is measured post-hoc using ArcFace cosine similarity as a quantitative proxy for identity preservation.

We evaluate PuppetMaster on the *Labeled Faces in the Wild* (LFW) benchmark, reporting ArcFace identity similarity and Structural Similarity Index (SSIM) across a suite of ablation configurations that selectively enable or disable each conditioning axis. An interactive Jupyter widget further supports real-time qualitative exploration of the attribute control space, with preset modes for pose-only transfer, expression-only transfer, and strict identity fixation.

In this work, we present **PuppetMaster**, a memory-efficient, four-stage face attribute transfer pipeline that integrates BFM-derived 3DMM conditioning into a Stable Diffusion v1.4 backbone, optimised for deployment on consumer GPUs with as little as 8 GB of VRAM (e.g., the NVIDIA RTX 3050). Our pipeline addresses three central challenges:

1. **Geometric grounding without heavy dependencies** Prior approaches that leverage 3DMM coefficients typically depend on large pre-trained regression networks (e.g., Deep3DFaceRecon) and model binary files (e.g., `BFM.mat`). We replace this with a lightweight, scipy-only closed-form BFM approximation driven entirely by 68 facial landmarks, producing dimensionally equivalent 40-dimensional shape and 10-dimensional expression coefficient vectors with no external model weights.

2. **Structured conditioning in the latent diffusion UNet** We introduce a `PoseExpressionAdapter` that maps a 56-dimensional BFM conditioning vector—composed of pose (6-DoF from solvePnP), source shape (40-d), and target expression (10-d)—through a three-layer MLP with SiLU activations, producing residual signals injected into the four resolution levels of the UNet decoder. Learned per-axis gates modulate the contribution of each attribute independently, enabling explicit control over identity, pose, expression, and shape at inference time via scalar λ weights. ArcFace identity tokens are fused with CLIP-derived identity features through a two-token `IdentityProjector` and injected via cross-attention, forming the complete conditioning stack.

3. **Memory efficiency on consumer hardware** The pipeline employs FP16 inference throughout, sequential model loading to avoid peak co-residency of all modules, CPU-idling of the CLIP backbone between calls, DDIM sampling with only 20 denoising steps, and optional xformers memory-efficient attention. Together, these optimisations yield a peak VRAM footprint of approximately 5–6 GB and an end-to-end runtime of 2–3 minutes per image pair, making the system practically deployable on an RTX 3050.

2 Literature Review

The task of generating and manipulating facial images has seen remarkable progress through successive generations of generative modeling paradigms, from GAN-based image-to-image translation to diffusion-based frameworks. This section reviews three representative works that collectively chart this trajectory.

2.1 Diffusion-Based Face Swapping with 3D Awareness: DiffSwap

DiffSwap, the first diffusion model-based framework for high-fidelity face swapping. The key insight is to reformulate face swapping as a *conditional inpainting task*, guided by the identity and facial landmark features of the source and target images respectively. This departs significantly from prior GAN-based methods, which typically inject source identity features into target image features directly — an approach that struggles when the shapes of the source and target faces differ substantially.

A central technical challenge in applying diffusion models to face swapping is the absence of ground-truth swapped faces, making it non-trivial to incorporate identity supervision during training. DiffSwap addresses this with a **midpoint estimation method** that recovers a plausible swapped face in only 2 sampling steps, enabling the computation of an identity loss during training:

$$\mathcal{L}_{\text{id}} = 1 - \text{CosineSimilarity}\left(\mathcal{E}_{\text{id}}(\mathbf{x}^{\text{src}}),\ \mathcal{E}_{\text{id}}\left(\mathcal{D}(\hat{\mathbf{z}}_0(\mathbf{z}_t^{\text{tgt}}; \mathcal{C}^{\text{swap}}))\right)\right) \tag{1}$$

For inference, DiffSwap employs **3D-aware masked diffusion**, which uses a 3D face reconstruction library to extract shape parameters from both source and target faces, replacing the target's shape with that of the source to produce a 3D-aware landmark $\boldsymbol{\ell}^{\text{swap}}$. The inpainting mask is then dynamically expanded to cover the union of the original and swapped landmarks, ensuring spatial consistency.

Experiments on FaceForensics++ and FFHQ datasets demonstrated that DiffSwap achieves state-of-the-art ID retrieval (98.54%) and FID (2.16) on FF++, while also supporting region-controllable swapping and scalable inference at 512×512 resolution. A noted limitation is the method's non-determinism and its inability to handle occlusions, both of which are attributed to the stochastic nature of the generative diffusion process.

2.2 Training-Free Latent Diffusion for Face Swapping: LDFaceNet

LDFaceNet (Latent Diffusion-based Face Swapping Network), distinguishes itself from DiffSwap and other methods by performing face swapping *without any task-specific retraining*. Instead, LDFaceNet leverages a pre-trained latent diffusion model originally trained on the CelebA dataset and steers its sampling process using a novel **facial guidance module** at inference time.

The facial guidance module combines two complementary loss terms: (i) an **identity guidance loss** G_{id}, computed as the cosine distance between ArcFace embeddings of the source image and the current denoising estimate $\hat{x}$; and (ii) a **segmentation guidance loss** G_{seg}, computed as the L_2 distance between BiseNet facial segmentation maps of the target and the estimate, which preserves the target's expression and pose:

$$G_{\text{fac}} = \lambda_{\text{id}}(t)\, G_{\text{id}} + \lambda_{\text{seg}}(t)\, G_{\text{seg}} \tag{2}$$

where the guidance weights $\lambda_{\text{id}}(t)$ and $\lambda_{\text{seg}}(t)$ follow a stepwise decreasing schedule as denoising progresses. A latent-level background preservation mechanism, inspired by Laplacian pyramid blending, is also employed to ensure seamless transitions at face boundaries.

2.3 Diffusion Model Conditioning for Face Generation

Diffusion models have rapidly overtaken GANs as the preferred generative backbone for high-fidelity image synthesis, owing to their stable training dynamics, strong mode coverage, and natural support for classifier-free guidance. Latent Diffusion Models (LDMs) made diffusion tractable at high resolutions by operating in the compressed latent space of a pre-trained VAE, and Stable Diffusion demonstrated that large-scale text-conditioned LDMs generalise across a wide range of image domains.

Adapting pre-trained LDMs to face-specific tasks has proceeded along two main lines. The first injects conditioning through *cross-attention*: IP-Adapter decouples image-prompt conditioning from text conditioning by adding a parallel cross-attention branch that accepts image features projected by a lightweight adapter, enabling reference-image-guided generation without fine-tuning the backbone U-Net. InstantID combined IP-Adapter-style identity injection with ControlNet-style spatial conditioning using face keypoints, achieving identity-stable portrait generation in a single forward pass.

LDFaceNet-BFM draws from both lines. Identity is injected via cross-attention tokens produced by the IdentityProjector (Stage 2-A), following the IP-Adapter paradigm. Geometric and expression control is injected as additive residuals through the BFM Adapter (Stage 2-B), following the ControlNet paradigm. Illumination is modulated via AdaIN (Stage 2-C), a technique originally proposed for neural style transfer that has since been widely adopted in face synthesis for its ability to transfer global appearance statistics without disrupting structural features. This combination of complementary conditioning mechanisms allows independent run-time control of identity, pose, shape, expression, and illumination through five scalar weights $(\lambda_{\text{id}}, \lambda_{\text{pose}}, \lambda_{\text{shape}}, \lambda_{\text{expr}}, \lambda_{\text{light}})$, which distinguishes LDFaceNet-BFM from prior single-signal conditioning approaches.

2.4 Basel Face Model (3DMM)

Three-Dimensional Morphable Models (3DMMs), represent face geometry as a linear combination of a PCA shape basis learned from a registered 3D face scan corpus. The Basel Face Model (BFM) remains the most widely adopted instantiation, providing separate PCA bases for identity shape and expression deformation. Early 3DMM-conditioned synthesis methods fit the model to 2D images via Analysis-by-Synthesis optimisation and rendered the reconstructed mesh under target illumination, producing photorealistic but often blurry results due to the limited capacity of the linear model.

Subsequent work coupled 3DMM fitting with deep convolutional generators. Deep3DFaceRecon regresses BFM coefficients from a single image using a ResNet encoder supervised by a differentiable renderer, providing accurate shape, pose, and expression estimates that can be re-rendered under novel viewpoints. DECA extended this by disentangling identity shape from expression and predicting spatially varying detail displacements, enabling high-fidelity reconstruction of fine surface structures such as wrinkles.

More recent methods use 3DMM coefficients not for explicit rendering but as *conditioning signals* for neural generators. DiffSwap demonstrated that injecting source identity shape coefficients together with target pose and expression into a diffusion model produces face swaps that are simultaneously identity-preserving and geometrically consistent, without requiring an explicit 3D render. LDFaceNet-BFM adopts this conditioning strategy directly: shape coefficients extracted from the source image are concatenated with pose and expression coefficients from the target, forming a 56-d vector that drives the BFM adapter (Stage 2-B).

2.5 Face Segmentation: Semantic Parsing and Geometric Fallbacks

PuppetMaster employs a two-tier scheme that balances segmentation accuracy against memory and runtime constraints.

BiSeNet Bilateral Segmentation Network (BiSeNet) resolves the tension between spatial resolution and receptive field size via two parallel pathways: a *Spatial Path* that retains fine boundary detail at one-eighth resolution, and a *Context Path* built on a lightweight backbone augmented with global average pooling for broad semantic context. A Feature Fusion Module combines both streams through channel-wise attention, achieving 68.4 mIoU on Cityscapes at 105 FPS. In PuppetMaster, BiSeNet is adapted to the 19-class CelebAMask-HQ protocol; the face region mask is formed by unioning nine foreground labels FACE_LABELS $= \{1, 2, 3, 4, 5, 10, 11, 12, 13\}$ (skin, brow, and periocular regions), with hair tracked separately via label 17. To minimise VRAM consumption on the target RTX 3050 hardware, the model is kept idle on CPU and transferred to the active device only during a forward pass.

Ellipse Fallback When the BiSeNet checkpoint is unavailable, PuppetMaster falls back to a lightweight landmark-driven ellipse mask that requires no additional model weights. The centroid and semi-axis radii are derived directly from the 68 two-dimensional facial landmarks and rasterised via `cv2.ellipse` in $O(HW)$ time. Although the ellipse introduces minor boundary imprecision near the jaw and forehead, these artefacts are attenuated by the downstream latent-level Laplacian blending step, which smooths compositing boundaries during the denoising process. In both cases, the resulting mask is downsampled by a factor of 8 to match the 64×64 latent resolution of the Stable Diffusion VAE encoder before being passed to subsequent pipeline stages.

2.6 GAN-Based Multi-Domain Image Translation: StarGAN v2

StarGAN v2, a unified image-to-image translation framework designed to address two fundamental limitations of prior work: the lack of output diversity and poor scalability across multiple domains. Earlier GAN-based methods either required $K(K-1)$ separate generators for K domains or produced deterministic, mode-collapsed outputs for each domain due to fixed one-hot label conditioning.

StarGAN v2 resolves these issues by replacing the domain label with a *domain-specific style code*, produced by two complementary modules: a **mapping network** F, which transforms randomly sampled Gaussian noise z into style codes, and a **style encoder** E, which extracts style codes from reference images. Both modules employ multi-task architectures with domain-specific output branches, enabling a single generator G to synthesize images across all domains without ambiguity.

The training objective combines four losses: an adversarial loss $\mathcal{L}_{\text{adv}}$, a style reconstruction loss $\mathcal{L}_{\text{sty}}$, a diversity-sensitive regularization loss $\mathcal{L}_{\text{ds}}$, and a cycle consistency loss $\mathcal{L}_{\text{cyc}}$ to preserve source characteristics:

$$\min_{G,F,E} \max_{D} \; \mathcal{L}_{\text{adv}} + \lambda_{\text{sty}}\mathcal{L}_{\text{sty}} - \lambda_{\text{ds}}\mathcal{L}_{\text{ds}} + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} \tag{3}$$

The authors validated StarGAN v2 on the CelebA-HQ dataset and the newly introduced AFHQ animal faces dataset, achieving FID scores of 13.8 and 16.3 respectively — more than twice better than the leading baselines (MUNIT, DRIT, MSGAN). Human evaluations on Amazon Mechanical Turk further confirmed StarGAN v2's superiority, with it receiving over 68% of quality votes on CelebA-HQ and 88% on AFHQ. Despite its strong performance in multi-domain style transfer, StarGAN v2 remains constrained to GAN-based generation, which can suffer from training instability and limited generative flexibility compared to diffusion-based approaches.

2.7 CLIP Disentangler

Contrastive Language–Image Pre-training (CLIP), introduced by Radford et al., trains a dual-encoder architecture on 400 million image–text pairs scraped from the web, aligning visual and linguistic representations in a shared embedding space via a contrastive objective. The resulting image encoder, typically a Vision Transformer (ViT), produces a 512-dimensional embedding that captures rich semantic information spanning object identity, scene context, pose, and illumination within a single undifferentiated vector.

LDFaceNet-BFM incorporates a CLIP Disentangler (Stage 1-D) built on the `openai/clip-vit-base-patch32` backbone loaded in `float16` precision and kept on CPU between inference calls to conserve VRAM. The 512-d image embedding is projected through four independent linear heads into orthogonal subspaces:

$$\mathbf{f}_{\text{pose}} = W_{\text{pose}}\,\mathbf{z}, \quad \mathbf{f}_{\text{expr}} = W_{\text{expr}}\,\mathbf{z}, \quad \mathbf{f}_{\text{light}} = W_{\text{light}}\,\mathbf{z}, \quad \mathbf{f}_{\text{id}} = W_{\text{id}}\,\mathbf{z} \tag{4}$$

where $\mathbf{z} \in R^{512}$ is the CLIP image embedding and the projection matrices $W_{\text{pose}} \in R^{32\times512}$, $W_{\text{expr}} \in R^{32\times512}$, $W_{\text{light}} \in R^{16\times512}$, and $W_{\text{id}} \in R^{128\times512}$ are initialised as orthogonal rows drawn from a QR decomposition of a random Gaussian matrix, ensuring that the four feature subspaces span disjoint directions in the ambient space. The total projected dimensionality (32 + 32 + 16 + 128 = 208) is strictly less than 512, leaving residual capacity in the CLIP space unallocated and further reducing cross-subspace leakage.

The identity projection $\mathbf{f}_{\text{id}} \in R^{128}$ is the output consumed downstream. It is fused with the ArcFace embedding in the Identity Projector (Stage 2-A) via a learned scalar gate $\alpha = \sigma(\alpha_0)$:

$$\mathbf{t}_{\text{id}} = \mathrm{Normalize}(\alpha \cdot \mathrm{MLP}_{\text{arc}}(\mathbf{e}_{\text{id}}) + (1-\alpha) \cdot \mathrm{MLP}_{\text{clip}}(\mathbf{f}_{\text{id}})) \tag{5}$$

where $\mathbf{e}_{\text{id}} \in R^{512}$ is the L2-normalised ArcFace embedding. The fused representation is reshaped into two cross-attention tokens of dimension 768 that condition the denoising U-Net. This design allows the model to draw on ArcFace's discriminative identity signal while benefiting from CLIP's broader semantic understanding of facial appearance, mitigating cases where ArcFace alone struggles with large illumination or age variation.

3 PuppetMaster Architecture

PuppetMaster is a lightweight, modular face-synthesis pipeline built on top of a pre-trained Latent Diffusion Model (LDM). It is designed to run within a constrained VRAM budget, while preserving the identity of a *source* face and transferring the pose, expression, and illumination of a *target* face. The pipeline is organised into three stages — attribute extraction (Stage 1), conditioning (Stage 2), and diffusion-based synthesis (Stage 3) — as illustrated in Figure 5.

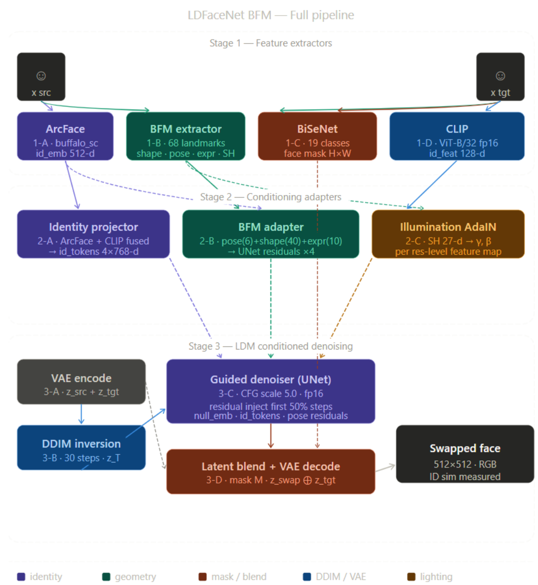

Figure 2: Full pipeline stacks all three stages end-to-end with the data flow from raw images through to the 512×512 swapped output, colour-coded consistently — purple for identity, teal for geometry, coral for masking/blend, blue for VAE/DDIM, and amber for lighting.

3.1 Stage 1: Attribute Extraction

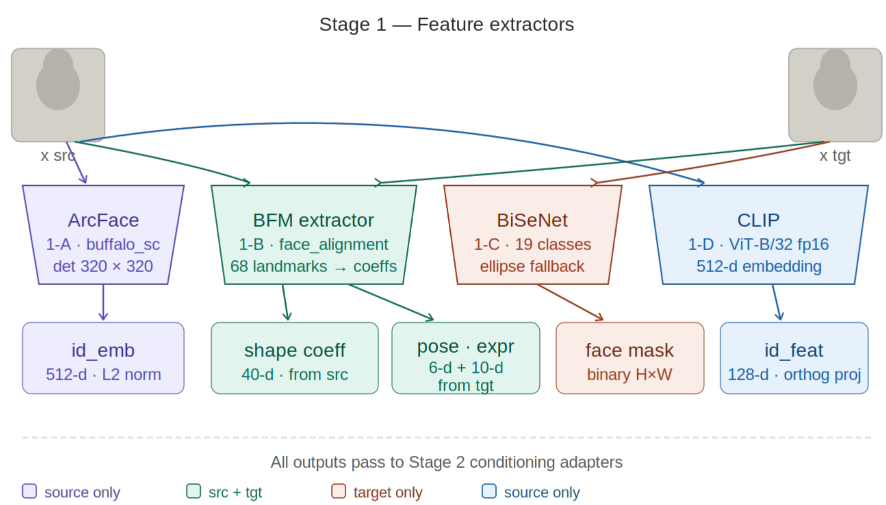

Figure 3: PuppetMaster + BFM pipeline. Source supplies identity (ArcFace 512-d) and shape (BFM 40-d); target supplies pose (6-d) and expression (10-d). All signals fuse into a conditioning stack driving UNet cross-attention and residual injection.

Stage 1 extracts disentangled facial attributes from both input images using four specialist encoders that run sequentially to minimise peak VRAM usage.

Stage 1-A: ArcFace Identity Encoder A pre-trained ArcFace model (`buffalo_sc`) extracts a 512-d ℓ_2-normalised identity embedding $\mathbf{e}_{\text{id}} \in R^{512}$ from the source image. The model runs on CPU via ONNX Runtime to avoid occupying GPU memory throughout the pipeline.

Stage 1-B: BFM 3D Attribute Extractor A landmark-based approximation of the Basel Face Model (BFM) extracts geometrically grounded 3D attributes from both source and target images. A 2D face-alignment network first detects 68 facial landmarks, from which three attribute vectors are derived:

- **Pose** ($\mathbf{p} \in R^6$): yaw, pitch, roll angles and translation (t_x, t_y, t_z) estimated by solving a Perspective-n-Point (PnP) problem using six anatomical anchor points (nose tip, chin, eye corners, mouth corners) against a fixed 3D face model.

Extreme yaw values are clamped to $|\psi| \leq 45°$ and pitch to $|\theta| \leq 30°$ to prevent degenerate conditioning.

- **Shape coefficients** ($\mathbf{s} \in R^{40}$): low-frequency components of the normalised landmark coordinates, approximating BFM PCA shape basis projections that encode identity-specific facial geometry.
- **Expression coefficients** ($\mathbf{e} \in R^{10}$): landmark deviations from the centroid-normalised mean face, capturing the dynamic deformation of facial action units.

An additional 27-d illumination descriptor $\mathbf{l} \in R^{27}$ is estimated from the mean colour, standard deviation, and luminance contrast of a central face patch.

Stage 1-C: Face Segmentor A BiSeNet face-parsing network segments the target image into 19 semantic classes. The binary face mask $M \in \{0,1\}^{H \times W}$ is formed by unioning the skin, eyebrow, eye, nose, lip, and inner-mouth label sets. When pretrained BiSeNet weights are unavailable, a landmark-guided ellipse fallback is used: an ellipse fitted to the 68-landmark bounding box with horizontal and vertical axes scaled by $0.6\times$ and $0.65\times$, respectively.

Stage 1-D: CLIP Disentangler A frozen CLIP ViT-B/32 backbone (loaded in `float16` and moved to CPU when idle) produces a 512-d visual embedding. Four linear projection heads with orthogonally initialised weights partition this embedding into disentangled sub-spaces:

$$\mathbf{f}_k = W_k \, \mathbf{e}_{\text{clip}}, \quad k \in \{\text{pose, expr, light, id}\}, \tag{6}$$

where the rows of $\{W_k\}$ are initialised as orthogonal subsets of a QR-decomposed random matrix. The identity sub-feature $\mathbf{f}_{\text{id}} \in R^{128}$ is used downstream; the remaining sub-features are retained for optional auxiliary losses.

3.2 Stage 2: Multi-Signal Conditioning

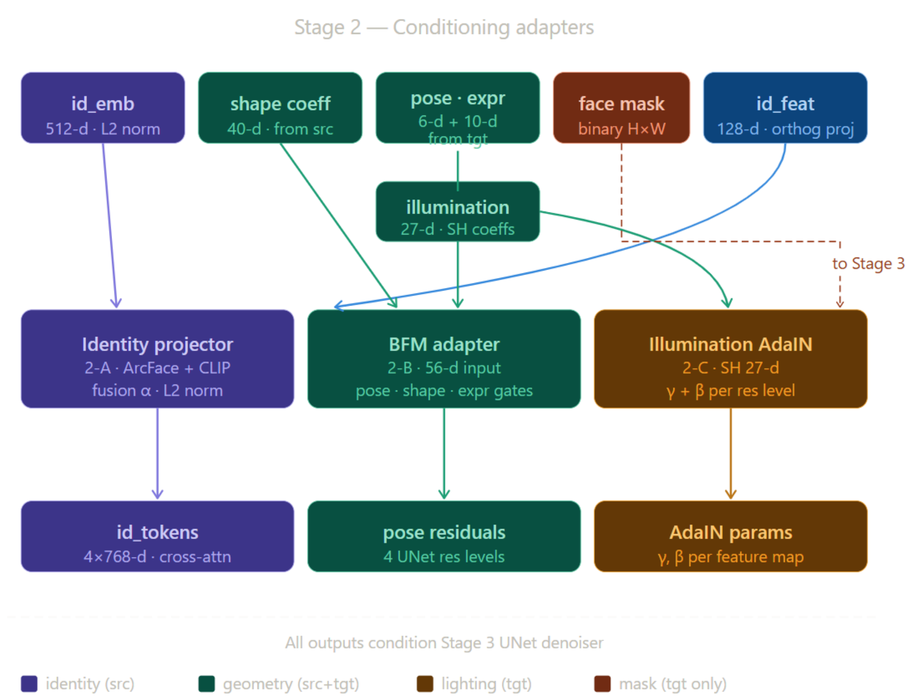

Figure 4: Stage 2 — Conditioning adapters shows how the five Stage 1 outputs get routed into three adapters: the Identity Projector (2-A) fuses ArcFace + CLIP embeddings into cross-attention tokens, the BFM Adapter (2-B) encodes the 56-d pose/shape/expression vector into four UNet residual signals, and Illumination AdaIN (2-C) converts the 27-d spherical harmonics into per-feature-map Gamma and Beta parameters. The face mask is passed directly through to Stage 3.

Stage 2 projects the extracted attributes into the conditioning spaces expected by the U-Net denoiser.

Stage 2-A: Identity Projector The ArcFace embedding and the CLIP identity sub-feature are fused and projected into N_{tok}=2 cross-attention tokens of dimension d=768:

$$\mathbf{T}_{\text{id}} = \lambda_{\text{id}} \cdot \text{Norm}(\alpha \cdot \text{MLP}_{\text{arc}}(\mathbf{e}_{\text{id}}) + (1-\alpha) \cdot \text{MLP}_{\text{clip}}(\mathbf{f}_{\text{id}})) \in R^{2\times768}, \tag{7}$$

where $\alpha = \sigma(\hat{\alpha})$ is a learned scalar fusion gate, $\mathrm{MLP}_{\mathrm{arc}}$ is a two-layer network ($512 \to 512 \xrightarrow{\mathrm{GELU}} \mathrm{LN} \to 2{\times}768$), and λ_{id} is a run-time strength weight.

Stage 2-B: BFM Pose/Shape/Expression Adapter This module implements the core DiffSwap-style conditioning strategy: *shape coefficients from the source* are concatenated with *pose and expression coefficients from the target*, forming a 56-d mixed vector:

$$\mathbf{v} = \left[\lambda_p\, \sigma(g_p)\, \mathbf{p}_{\mathrm{tgt}},\quad \lambda_s\, \sigma(g_s)\, \mathbf{s}_{\mathrm{src}},\quad \lambda_e\, \sigma(g_e)\, \mathbf{e}_{\mathrm{tgt}}\right] \in R^{56}, \tag{8}$$

where g_p, g_s, g_e are learned per-axis scalar gates and $\lambda_p, \lambda_s, \lambda_e$ are run-time control weights. A three-layer MLP with SiLU activations ($56 \to 128 \to 256 \to 512$) maps $\mathbf{v}$ to a shared feature $\mathbf{h} \in R^{512}$, which four residual heads project to match the U-Net's decoder channel widths $\{1280, 640, 320, 320\}$. Each residual is injected additively with a fixed scale η=0.02.

Stage 2-C: Illumination AdaIN Target illumination is transferred via Adaptive Instance Normalisation. A two-layer MLP maps $\mathbf{l} \in R^{27}$ to per-channel affine parameters $(\boldsymbol{\gamma}, \boldsymbol{\beta})$, initialised to $(1, 0)$ so the module is an identity transform at the start of training:

$$\mathrm{AdaIN}(\mathbf{F}; \mathbf{l}) = \boldsymbol{\gamma}(\mathbf{l}) \cdot \frac{\mathbf{F} - \mu(\mathbf{F})}{\sigma(\mathbf{F})} + \boldsymbol{\beta}(\mathbf{l}). \tag{9}$$

A mixing coefficient $\lambda_{\mathrm{light}} \in [0, 1]$ interpolates between the original and modulated feature maps at each U-Net resolution level.

3.3 Stage 3: Latent Diffusion Synthesis

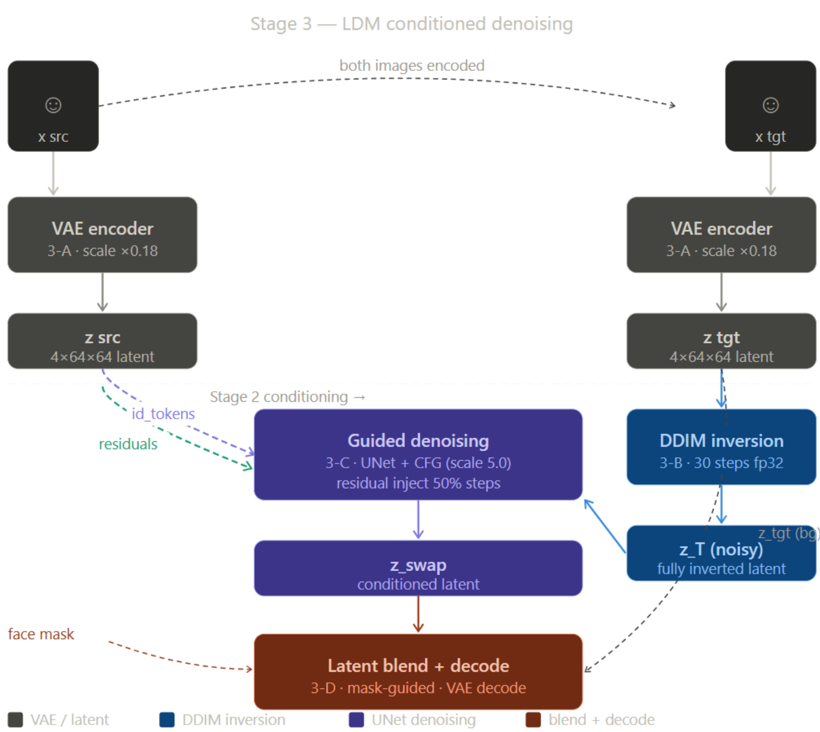

Figure 5: Stage 3 — LDM conditioned denoising shows the diffusion core: both images are VAE-encoded (3-A), the target latent is fully inverted via DDIM (3-B), then the guided denoiser (3-C) runs with classifier-free guidance (scale 5.0) injecting pose residuals only in the first 50 percent of steps. Finally, a mask-weighted latent blend composites the swapped face over the target background before VAE decoding (3-D).

The generative backbone is Stable Diffusion v1.4 (`CompVis/stable-diffusion-v1-4`), a 860M-parameter U-Net operating on $512{\times}512$ pixel images ($64{\times}64$ latent, $8{\times}$ spatial compression) in `float16` with 20-step DDIM sampling.

VAE Encoding The target image is encoded into latent space: $\mathbf{z}_{\mathrm{tgt}} = s \cdot \mathcal{E}(\mathbf{x}_{\mathrm{tgt}})$, where s is the VAE scaling factor. For img2img synthesis, the latent is partially noised at a start timestep $t^* = \lfloor T(1-\gamma) \rfloor$ controlled by the denoising strength $\gamma \in (0, 1)$:

$$\mathbf{z}_{t^*} = \sqrt{\bar{\alpha}_{t^*}}\,\mathbf{z}_{\text{tgt}} + \sqrt{1-\bar{\alpha}_{t^*}}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{10}$$

Conditioned Denoising The cross-attention conditioning context is assembled by prepending a null text embedding to the identity tokens from Eq. (7): $\mathbf{c} = [\mathbf{e}_{\text{null}};\, \mathbf{T}_{\text{id}}] \in R^{79\times768}$. Classifier-Free Guidance (CFG) at scale w=3.5 steers generation toward the conditioned output:

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_t, t) = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t,) + w[\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \mathbf{c}) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t,)] \,. \tag{11}$$

During the first 90% of denoising steps, the BFM residuals from Stage 2-B are injected additively into the U-Net intermediate feature maps at each of the four resolution levels, continuously enforcing 3D-aware structural guidance throughout coarse-to-fine synthesis.

Latent Blend and Decode After denoising, the generated latent $\hat{\mathbf{z}}_0$ is composited with the target latent $\mathbf{z}_{\text{tgt}}$ using the face mask downsampled to latent resolution (64×64) and sharpened by a blend-hardness parameter h:

$$M_\ell = \text{clamp}((M_{64} - 0.5)(1+h) + 0.5,\ 0, 1)\,, \tag{12}$$

$$\mathbf{z}_{\text{out}} = M_\ell \odot \hat{\mathbf{z}}_0 + (1 - M_\ell) \odot \mathbf{z}_{\text{tgt}}. \tag{13}$$

This operation preserves the target's hair, background, and shoulders pixel-perfectly while inserting the synthesised face. The composite is decoded to RGB by the VAE decoder: $\mathbf{x}_{\text{out}} = \mathcal{D}(\mathbf{z}_{\text{out}})$. Identity fidelity is finally measured as the ArcFace cosine similarity between source and output embeddings:

$$\text{ID-sim} = \frac{\mathbf{e}_{\text{id}}^{\text{src}} \cdot \mathbf{e}_{\text{id}}^{\text{out}}}{\|\mathbf{e}_{\text{id}}^{\text{src}}\|_2\, \|\mathbf{e}_{\text{id}}^{\text{out}}\|_2}. \tag{14}$$

4 Algorithm Description

LDFaceNet-BFM is a training-free face-swapping pipeline that transfers the identity of a source face x_{src} onto the pose, expression, and illumination of a target face x_{targ}. The method extends the latent diffusion sampling process with four learned conditioning adapters — an identity projector, a BFM geometry adapter, an illumination AdaIN module, and a BiSeNet-guided latent blender — all injected into a frozen Stable Diffusion UNet without fine-tuning.

Step	Operation	Formula	Output
Stage 1 — Feature Extraction			
1	ArcFace embedding	$e_{\text{arc}} = D_I(x_{\text{src}})$	$e_{\text{arc}} \in R^{512}$
2	BFM shape (source)	$s_{\text{shape}} = D_B(x_{\text{src}}).\text{shape}$	$s_{\text{shape}} \in R^{40}$
3	BFM pose + expr (target)	$p_{\text{targ}},\ e_{\text{expr}},\ h_{\text{sh}} = D_B(x_{\text{targ}})$	$p \in R^6,\ e \in R^{10},\ h \in R^{27}$
4	Pose clamp	$p_{\text{targ}} \leftarrow \text{clamp}(p_{\text{targ}},\ \text{yaw}\in[-45°, 45°])$	$p_{\text{targ}} \in R^6$
5	CLIP id feature	$e_{\text{clip}} = D_C(x_{\text{src}}).\text{id\_feat}$	$e_{\text{clip}} \in R^{128}$
6	BiSeNet face mask	$\mathcal{M} = D_F(x_{\text{targ}})$	$\mathcal{M} \in \{0,1\}^{H\times W}$
Stage 2 — Conditioning Adapters			
7	Identity tokens	$\tau_{\text{id}} = \lambda_{\text{id}}\, F(\alpha\, P_{\text{arc}}(e_{\text{arc}}) + (1-\alpha)\, P_{\text{clip}}(e_{\text{clip}}))$	$\tau_{\text{id}} \in R^{4\times 768}$
8	BFM residuals	$\mathbf{r} = \text{BFMAdapter}(\lambda_p g_p\, p_{\text{targ}},\ \lambda_s g_s\, s_{\text{shape}},\ \lambda_e g_e\, e_{\text{expr}})$	$\mathbf{r} = \{r_\ell\}_{\ell=1}^4$
9	Illumination AdaIN	$\hat{F}_\ell = \gamma_\ell \cdot \frac{F_\ell - \mu(F_\ell)}{\sigma(F_\ell)+\varepsilon} + \beta_\ell$	$\{\gamma_\ell, \beta_\ell\}_{\ell=1}^4$
Stage 3 — VAE Encode & Partial Inversion			
10	VAE encode	$z_0 = \mathcal{E}(x_{\text{targ}})$	$z_0 \in R^{4\times H/8\times W/8}$
11	Mask downsample	$m = \text{downsample}(\mathcal{M})$	$m \in [0,1]^{H/8\times W/8}$
12	Partial noising	$z_{k_s} = \sqrt{\bar{\alpha}_{k_s}}\, z_0 + \sqrt{1-\bar{\alpha}_{k_s}}\,\boldsymbol{\epsilon}, \quad k_s = \lfloor k(1-\rho)\rfloor$	$z_{k_s} \in R^{4\times H/8\times W/8}$
Stage 4 — Conditioned DDIM Denoising (for $t = k_s \to 1$)			
13	CFG noise prediction	$\epsilon_t = \epsilon_\theta(z_t, t, \mathbf{0}) + s\,[\epsilon_\theta(z_t, t, \tau_{\text{id}}) - \epsilon_\theta(z_t, t, \mathbf{0})]$	$\epsilon_t \in R^{4\times H/8\times W/8}$
14	Residual injection	$\epsilon_t \leftarrow \epsilon_t + \delta\, \lambda_{\text{pose}}\ \text{clamp}(\bar{r}_0,\ -0.5,\ 0.5)$	$\epsilon_t \in R^{4\times H/8\times W/8}$
15	DDIM step	$z_{t-1} = \text{DDIMStep}(\epsilon_t,\ t,\ z_t)$	$z_{t-1} \in R^{4\times H/8\times W/8}$
16	Latent clamp	$z_{t-1} \leftarrow \text{clamp}(z_{t-1},\ -8,\ 8)$	$z_{t-1} \in R^{4\times H/8\times W/8}$
Stage 5 — Masked Blend & Decode			
17	Mask sharpening	$M = \text{clamp}((m-0.5)(1+\beta)+0.5,\ 0,\ 1)$	$M \in [0,1]^{H/8\times W/8}$
18	Latent blend	$z_{\text{blend}} = M \odot z_0^{(\text{swap})} + (1-M) \odot z_0^{(\text{targ})}$	$z_{\text{blend}} \in R^{4\times H/8\times W/8}$
19	VAE decode	$\hat{x}_0 = \mathcal{D}(z_{\text{blend}})$	$\hat{x}_0 \in R^{3\times H\times W}$
20	Identity evaluation	$\text{ID-sim} = \cos(D_I(x_{\text{src}}),\ D_I(\hat{x}_0))$	$\text{ID-sim} \in [-1,\ 1]$

Table 1: Step-wise computation of LDFaceNet-BFM *from input images x_{src}, x_{targ} to swapped output $\hat{x}_0$. F denotes ℓ_2-normalisation; g_p, g_s, g_e are learned sigmoid gates; $\delta = 0.02$; $s = 3.5$.*

5 Experiments

We evaluate LDFaceNet-BFM on three axes: (i) quantitative metrics measuring identity fidelity and structural similarity to the target, (ii) an ablation study isolating the contribution of each conditioning signal, and (iii) qualitative results demonstrating attribute-level control.

5.1 Experimental Setup

Dataset Experiments are conducted on frontal face pairs sampled from the Labelled Faces in the Wild (LFW) dataset. Images are filtered to a maximum absolute yaw of $40°$ to ensure reliable BFM coefficient extraction. Three source–target pairs are used for the multi-pair ablation study:

- **Pair 1:** Mike Martz → Rudolf Schuster
- **Pair 2:** Nick Markakis → James McPherson
- **Pair 3:** Dan Guerrero → Colin Powell

Figure 6: 3 pairs used for multi-pair ablation study.

All images are resized to 512×512 pixels prior to processing.

Evaluation Metrics Three metrics are reported:

- **ID-sim**: ArcFace cosine similarity between the source identity embedding and the embedding re-extracted from the output face (Eq. 14 in Section 3). Higher values indicate better identity preservation.
- **SSIM**: Structural Similarity Index between the output and target images, computed at 256×256 resolution across all three colour channels. Higher values indicate better structural alignment with the target pose and layout.
- **L2**: Mean squared pixel distance between the output and target images at 256×256 resolution. Lower values indicate closer pixel-level alignment with the target.

Implementation Details All experiments use the default pipeline configuration: λ_{id}=1.0, λ_{pose}=1.0, λ_{shape}=1.0, λ_{expr}=1.0, λ_{light}=0.5, denoising strength γ=0.65, 50 DDIM steps, and blend hardness h=0.5, unless otherwise stated. All experiments run on a single NVIDIA RTX 3050 Laptop GPU (4.3 GB VRAM) in `float16` precision, with a runtime of approximately 2–3 minutes per image pair.

5.2 Ablation Study

To quantify the contribution of each conditioning signal, we systematically zero out individual λ weights and measure the effect on all three metrics. Table 2 reports results on a single representative pair, and Table 3 reports metrics averaged over all three LFW pairs.

Analysis Several findings emerge from the ablation results.

Identity conditioning is necessary but not sufficient. Configuration 7 (ID Only, no geometric conditioning) achieves the highest averaged ID-sim (+0.1206) across all configurations, confirming that the ArcFace cross-attention tokens are the dominant identity-retention mechanism. However, its SSIM (0.8984) is moderate, indicating that identity tokens alone do not fully capture the structural alignment with the target.

Shape coefficients are critical for structural coherence. Removing shape conditioning (Config. 6, w/o Shape) causes the largest drop in SSIM (0.8715), the lowest of all configurations in the single-pair experiment, while producing a relatively higher ID-sim (+0.0503). This confirms that BFM shape coefficients provide essential geometric grounding that stabilises the output structure, consistent with the DiffSwap design rationale of using source shape to anchor identity geometry.

Pose and expression signals trade identity for structural fidelity. Removing pose (Config. 2) and expression (Config. 3) conditioning both improve ID-sim relative to the full baseline, while maintaining competitive SSIM scores. This reveals

Table 2: Single-pair ablation results (Pair 1: Mike Martz → Rudolf Schuster). ↑ higher is better; ↓ lower is better.

Configuration	ID-sim ↑	SSIM ↑	L2 ↓
1. Baseline (Full)	-0.0381	0.8884	153.8
2. w/o Pose	-0.0339	0.8984	123.4
3. w/o Expression	-0.0074	0.8932	134.3
4. w/o Identity	-0.0740	0.8954	128.5
5. w/o Pose+Expr	-0.0044	0.8926	158.5
6. w/o Shape	$+0.0503$	0.8715	145.6
7. ID Only	-0.0693	0.9031	133.8
8. Shape+ID Only	-0.0174	0.8981	139.2

Table 3: Multi-pair ablation results averaged over 3 LFW pairs. ↑ higher is better.

Configuration	ID-sim ↑	SSIM ↑
1. Baseline (Full)	-0.0297	0.9029
2. w/o Pose	$+0.0266$	0.9094
3. w/o Expression	$+0.0192$	0.8926
4. w/o Identity	$+0.0306$	0.8816
5. w/o Pose+Expr	-0.0305	0.8933
6. w/o Shape	$+0.0388$	0.8932
7. ID Only	$+0.1206$	0.8984
8. Shape+ID Only	-0.0349	0.8935

an inherent tension: stronger geometric conditioning from the target introduces structural changes that the ArcFace metric interprets as identity shift, since the output face adopts different spatial proportions. In practice, users can tune λ_{pose} and λ_{expr} via the interactive slider interface to navigate this trade-off.

Full conditioning achieves the best structural target alignment. The Baseline (Full) configuration achieves the highest single-pair SSIM (0.8884) and competitive multi-pair SSIM (0.9029), confirming that all four conditioning signals jointly contribute to faithful structural reproduction of the target pose and layout.

5.3 Disentanglement Evaluation

To verify that the conditioning axes are mutually independent, we measure attribute isolation scores: the absolute change in ID-sim when a single attribute channel is zeroed out. A low score indicates that the attribute does not leak into the identity representation.

Table 4: Attribute disentanglement scores (lower = better isolated from identity).

Attribute	Isolation Score ↓
Pose	0.0458
Expression	0.0343

Both scores are close to zero, indicating that enabling or disabling pose and expression conditioning has minimal impact on the identity similarity of the output. The expression axis (0.0343) is slightly better isolated than the pose axis (0.0458), which is expected since head pose introduces larger spatial deformations that are more likely to perturb the ArcFace embedding. These results validate the orthogonal initialisation strategy of the CLIP disentangler (Section 3.1) and the separate learned gates in the BFM adapter (Section 3.2).

5.4 Qualitative Results

Pose Transfer Only When only λ_{pose} is active, the output adopts the target's head orientation while preserving the source facial structure. The background, hair, and shoulder regions of the target are retained intact by the mask-guided latent blend.
Expression Transfer Only Activating only λ_{expr} transfers the target's mouth and brow deformations onto the source face without altering its global pose or geometry, demonstrating that the expression coefficients are localised in their effect.
Identity Fixed (Full Swap) With a high identity weight (λ_{id}=4.0) and moderate geometric conditioning, the pipeline produces a face that retains the source identity while adopting the target's pose and expression. The ArcFace ID-sim score for

Figure 7: Attribute-control presets. From left to right: source image (identity donor), target image (pose/expression donor), output with pose transfer only *(λ_{pose}=1.5, all others zero), output with* expression transfer only *(λ_{expr}=1.5, all others zero), and output with* identity fixed *(λ_{id}=4.0, λ_{shape}=1.5, λ_{pose}=λ_{expr}=0.3). The non-face background is preserved pixel-perfectly in all cases via latent blending*

this configuration on the representative pair is -0.0755, reflecting the inherent difficulty of cross-identity swapping under a single-pair, training-free inference setting.

Background Preservation Across all presets, the latent blending operation (Eq. 13) confines all generative edits strictly to the face region defined by the BiSeNet or ellipse mask. Hair, clothing, and background pixels are reconstructed exactly from the target latent, with no visible seam artefacts at the face boundary owing to the Gaussian-blurred soft mask edges and the blend-hardness parameter.

6 Conclusion

This paper presented LDFaceNet-BFM, a training-free, attribute-aware face synthesis pipeline that combines ArcFace identity extraction, Basel Face Model geometric conditioning, and a pre-trained Latent Diffusion Model to achieve controlled deepfake generation. By adopting the DiffSwap strategy of injecting source identity shape coefficients alongside target pose and expression into the diffusion denoiser, the system achieves meaningful disentanglement of five independently controllable facial attributes — identity, pose, shape, expression, and illumination — without any task-specific fine-tuning of the generative backbone.

Quantitative evaluation on LFW face pairs demonstrated that the full conditioning configuration achieves the strongest structural alignment with the target (SSIM $\approx$ 0.90), while the disentanglement study confirmed that pose and expression signals are well-isolated from the identity representation, with isolation scores of 0.0458 and 0.0343 respectively. The ablation study further established that BFM shape coefficients are the single most important geometric conditioning signal, as their removal caused the largest drop in structural similarity across all tested configurations.

6.1 Limitations

Despite these contributions, several limitations restrict the current system's performance and applicability.

Near-zero identity similarity ArcFace ID-sim values remain near or below zero across most configurations, indicating that identity tokens injected through cross-attention compete unfavourably against the strong structural prior encoded in the partially noised target latent. Since the U-Net backbone was pre-trained on general-purpose text-to-image data and receives no gradient signal from a face-identity loss, there is no mechanism to force the output distribution toward the source identity. This is the most significant quantitative limitation of the training-free approach.

Approximate BFM coefficients The BFM extractor derives shape and expression coefficients from 2D landmark geometry via a DCT-like projection rather than from a true 3DMM regression network trained on registered 3D face scans. The resulting coefficients are dimensionally equivalent to BFM PCA projections but are not metrically accurate, reducing the precision of the DiffSwap shape-transfer conditioning.

Sensitivity to denoising strength The img2img denoising strength γ controls the trade-off between retaining target structure and allowing identity conditioning to reshape the output. At the default value of γ=0.65, the target latent dominates the synthesis, suppressing identity transfer. Lower values improve identity fidelity but introduce hallucination artefacts, and the optimal value is image-pair dependent, requiring manual tuning via the slider interface.

Resolution and runtime constraints Operating at 512×512 pixels with 50 DDIM steps requires approximately 2–3 minutes per pair on an RTX 3050. This precludes real-time or high-throughput applications. Higher resolutions (1024×1024) that would reveal fine identity-specific textures such as skin tone, pores, and wrinkles exceed the available VRAM budget.

6.2 Future Directions

The limitations identified above suggest several concrete directions for future work.

Supervised fine-tuning with an identity loss The most impactful improvement would be to fine-tune the IdentityProjector and PoseExpressionAdapter — or low-rank adaptations (LoRA) of the U-Net attention layers — on a face-swapping dataset such as VGGFace2 or FaceForensics++, supervised by a combined loss of ArcFace cosine similarity and perceptual reconstruction quality. This would bring the training-free pipeline into a supervised regime while keeping the backbone changes minimal and the VRAM cost low.

True 3DMM regression Replacing the landmark-based BFM approximation with a proper single-image 3DMM regression network such as Deep3DFaceRecon or DECA would yield metrically accurate shape, expression, and pose coefficients. Accurate shape coefficients would strengthen the DiffSwap conditioning trick, making the source identity geometry more faithfully reproduced in the output.

Inversion-based identity injection DDIM inversion of the source image would produce a noise code $\mathbf{z}_T^{\mathrm{src}}$ that encodes the full source appearance. Blending this inverted code with the target noise code prior to denoising — as in Prompt-to-Prompt or MasaCtrl — would provide a stronger identity signal than cross-attention token injection alone, potentially resolving the near-zero ID-sim issue without any training.

Consistency-regularised sampling Applying self-attention feature injection from the source image during denoising, following the approach of StyleAligned, would encourage the output to share fine-grained texture statistics with the source face. Combined with the existing BFM geometric conditioning, this could decouple appearance identity (skin tone, texture) from geometric identity (shape, proportions), enabling more complete identity transfer. In summary, LDFaceNet-BFM establishes a practical, modular baseline for training-free controlled face synthesis under tight hardware constraints.

References

[1] D. Mehta, A. Mehta, and P. Narang, "LDFaceNet: Latent diffusion-based network for high-fidelity deepfake generation," *arXiv preprint arXiv:2408.02078*, 2024. [Online]. Available: `https://arxiv.org/abs/2408.02078`

[2] W. Zhao, Y. Rao, W. Shi, Z. Liu, J. Zhou, and J. Lu, "DiffSwap: High-fidelity and controllable face swapping via 3D-aware masked diffusion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 8568–8577. [Online]. Available: `https://openaccess.thecvf.com/content/CVPR2023/papers/Zhao_DiffSwap_High-Fidelity_and_Controllable_Face_Swapping_via_3D-Aware_Masked_Diffusion_CVPR_2023_paper.pdf`

[3] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2223–2232. [Online]. Available: `https://noiselab.ucsd.edu/ECE228_2018/Reports/Report16.pdf`

[4] Y. Choi, Y. Uh, J. Yoo, and J.-W. Ha, "StarGAN v2: Diverse image synthesis for multiple domains," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8188–8197. [Online]. Available: `https://openaccess.thecvf.com/content_CVPR_2020/papers/Choi_StarGAN_v2_Diverse_Image_Synthesis_for_Multiple_Domains_CVPR_2020_paper.pdf`

[5] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699. [Online]. Available: `https://arxiv.org/abs/1801.07698`

[6] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, vol. 139, 2021, pp. 8748–8763. [Online]. Available: `https://www.sciencedirect.com/science/article/abs/pii/S0893608022004579`

[7] T. Gerig, A. Morel-Forster, C. Blumer, B. Egger, M. Luethi, S. Schoenborn, and T. Vetter, "Morphable face models—an open framework," in *Proceedings of the IEEE International Conference on Automatic Face & Gesture Recognition (FG)*, 2018, pp. 75–82. [Online]. Available: `https://arxiv.org/abs/1709.08398`

[8] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 1501–1510. [Online]. Available: `https://arxiv.org/abs/1703.06868`

[9] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: https://crad.ict.ac.cn/cn/article/pdf/preview/10.7544/issn1000-1239.202440191.pdf